WORKED SOLUTION

# STEP II 1987, Question 3

Pure • Conjugate Surds and Integer Parts

**STEP II 1987, Question 3**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q3

## The key idea

Write $\alpha = a + \sqrt{a^2 - b}$ and $\beta = a - \sqrt{a^2 - b}$. These are conjugates: their sum $2a$ and product $b$ are integers, so $\alpha^n + \beta^n$ is always an integer. If the condition on $b$ forces the small conjugate $\beta$ to lie strictly between 0 and 1, then $\beta^n$ is the fractional part we need, and everything follows.

## Solution

Throughout, let

$$\alpha = a + \sqrt{a^2 - b}, \qquad \beta = a - \sqrt{a^2 - b}, \qquad \text{so} \qquad \alpha + \beta = 2a, \quad \alpha\beta = a^2 - (a^2 - b) = b.$$

**Where $\beta$ lies.** First, the square root is real. Since $b < 2a - 1$,

$$a^2 - b > a^2 - 2a + 1 = (a - 1)^2 \geq 0.$$

Taking positive square roots of this strict inequality gives $\sqrt{a^2 - b} > a - 1$. Also $b > 0$, so $a^2 - b < a^2$ and $\sqrt{a^2 - b} < a$. Combining these,

$$a - 1 < \sqrt{a^2 - b} < a \quad \Longrightarrow \quad 0 < \beta < 1.$$

(The condition is sharp: if $b = 2a - 1$ then $\sqrt{a^2 - b} = a - 1$ and $\beta = 1$ exactly.)

### Part (i)

Since $0 < \beta < 1$, we have $0 < \beta^n < 1$ for every positive integer $n$. Now $M$ is an integer and $\beta^n = M + s$ with $0 \leq s < 1$, so $M$ is the integer part of $\beta^n$, which is 0. Hence

$$M = 0 \qquad \text{and} \qquad s = \beta^n.$$

### Part (ii)

Let $S_n = \alpha^n + \beta^n$. We claim $S_n$ is an integer. Expanding both powers by the binomial theorem, with $d = \sqrt{a^2 - b}$,

$$S_n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}(d^k + (-d)^k) = 2 \sum_{k \text{ even}} \binom{n}{k} a^{n-k}(a^2 - b)^{k/2},$$

because the terms with odd $k$ cancel. Every surviving term is an integer, so $S_n \in \mathbb{Z}$. (Equivalently, $\alpha, \beta$ are roots of $t^2 - 2at + b = 0$, so $S_n = 2aS_{n-1} - bS_{n-2}$ with $S_0 = 2$ and $S_1 = 2a$.)

Therefore

$$\alpha^n = S_n - \beta^n, \qquad S_n \in \mathbb{Z}, \quad 0 < \beta^n < 1.$$

**Ask yourself** Why does this force $N = S_n$ and $r = \beta^n$?

The representation $\alpha^n = N - r$ with $N$ an integer and $0 \le r < 1$ is unique: $N$ must be the least integer that is at least $\alpha^n$. If $N' - r' = N - r$ with both $r, r' \in [0, 1)$, then $N - N' = r - r'$ is an integer strictly between $-1$ and $1$, so it is $0$. Our expression $S_n - \beta^n$ is of that form, so

$$N = S_n = \alpha^n + \beta^n, \qquad r = \beta^n = s.$$

### Part (iii)

Using $r = \beta^n$, $N = \alpha^n + \beta^n$ and $(\alpha\beta)^n = b^n$,

$$\begin{aligned} r^2 - Nr + b^n &= \beta^{2n} - (\alpha^n + \beta^n)\beta^n + \alpha^n\beta^n \\ &= \beta^{2n} - \alpha^n\beta^n - \beta^{2n} + \alpha^n\beta^n \\ &= 0, \end{aligned}$$

as required. In other words, $r$ is the smaller root of $t^2 - Nt + b^n = 0$, whose roots are $\alpha^n$ and $\beta^n$.

### The final part

Take $a = 8$ and $a^2 - b = 63$, so $b = 1$ and $\sqrt{63} = 3\sqrt{7}$. The condition $b < 2a - 1 = 15$ holds, so all of the above applies with $\alpha = 8 + 3\sqrt{7}$:

$$\left(8 + 3\sqrt{7}\right)^n = N - r, \qquad N \in \mathbb{Z},$$

and $\alpha^n$ falls short of the integer $N$ by $r$. Part (iii) with $b = 1$ gives $r(N - r) = 1$, that is,

$$r = \frac{1}{N - r} = \frac{1}{\left(8 + 3\sqrt{7}\right)^n} = \left(8 - 3\sqrt{7}\right)^n.$$

In particular $r \neq 0$, so the power is never itself an integer, and $r \to 0$. Now $3\sqrt{7} = \sqrt{63}$ is just below $\sqrt{64} = 8$, so

$$8 + 3\sqrt{7} \approx 16 = 2^4, \qquad \text{hence} \qquad r = \left(8 + 3\sqrt{7}\right)^{-n} \approx 16^{-n} = 2^{-4n}.$$

So for large $n$, $\left(8+3\sqrt{7}\right)^n$ is within about $2^{-4n}$ of an integer, lying just below it.

How good is "about"? Since $8+3\sqrt{7} \approx 15.937$, we have $r = 2^{-cn}$ with $c = \log_2(8+3\sqrt{7}) \approx 3.994$. The exponent agrees with $4n$ to within 0.2%, which is the sense in which the estimate holds. The ratio $r/2^{-4n} = (16/(8+3\sqrt{7}))^n$ creeps up slowly, from about 1.004 at $n=1$ to 1.04 at $n=10$.

## Checks

- $n=1$: $8+3\sqrt{7} \approx 15.93725$, which is $16-0.06275$, and $8-3\sqrt{7} \approx 0.06275$, as predicted. Also $r^2-16r+1=0$ has smaller root $8-\sqrt{63}$.
- $n=2$: $(8+3\sqrt{7})^2 = 127+48\sqrt{7} \approx 253.99606$, so $N = 254 = S_2 = 2\cdot 8\cdot 16 - 2$ and $r \approx 0.003937$, against $2^{-8} \approx 0.003906$.
- A computer check over thousands of random pairs $(a,b)$ with $b < 2a-1$ and $n \le 7$ confirmed $0 < \beta^n < 1$, $N - \alpha^n = \beta^n$ and $r^2 - Nr + b^n = 0$. A high-precision check for $a=8$, $b=1$, $n \le 10$ confirmed $N - \alpha^n = (8-3\sqrt{7})^n$; at $n=10$ this is $9.46\times 10^{-13}$, against $2^{-40} \approx 9.09\times 10^{-13}$.

## Common slips

- Showing only $\beta < 1$ and forgetting $\beta > 0$ (which needs $b > 0$). Both are needed for $M = 0$.
- Asserting $N = \alpha^n + \beta^n$ without saying why the integer and fractional parts in $N - r$ are unique.
- Expecting $r/2^{-4n} \to 1$. It doesn't: the approximation is in the exponent, since $8+3\sqrt{7}$ is close to 16 but not equal to it.

## Beyond STEP

The numbers $\alpha^n + \beta^n$ are an example of a trace: $\alpha$ is an algebraic integer whose conjugate lies inside the unit interval, so its powers get exponentially close to integers. Such numbers are called Pisot numbers; the golden ratio is another, which is why $\varphi^n$ is close to the Lucas number $L_n$. The case here is also tied to Pell's equation: $8^2 - 7\cdot 3^2 = 1$, so $(8,3)$ is the smallest positive solution of $x^2 - 7y^2 = 1$. Davenport's *The Higher Arithmetic* treats Pell's equation and continued fractions beautifully.

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